

# Phase 1: Business Foundation

## Healthcare Digital Platform — Business Analysis & Solution Design

**Subtitle**

Provider Lifecycle, Patient Care & Digital Healthcare Operations

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### Document Control

| Version | Date           | Author        | Description                                                |
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| 1.0     | June 2026      | Vaibhav Yadav | Initial Healthcare Case Study                              |
| 2.0     | September 2026 | Vaibhav Yadav | Enhanced Senior BA Documentation with Functional Artifacts |

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## **1. Executive Summary**

### **Project Background**

This case study presents the business analysis and solution design of a digital healthcare platform covering provider onboarding, verification, activation, and downstream healthcare operations.

The platform addresses challenges such as manual provider verification, multiple operational handoffs, limited process visibility, and delays across interconnected provider and patient processes. The proposed solution standardizes these processes through digital workflows, automated verification and business rules, exception handling, integrated services, notifications, SLA monitoring, and operational reporting.

Once a provider is verified and activated, the platform supports downstream activities including appointment management, patient consultation, and payment processing. Operational teams can monitor application status, SLA performance, and key business metrics through centralized dashboards.

The case study covers requirements analysis, process modelling, functional requirements, business rules, solution architecture, data modelling, API specifications, Agile user stories, acceptance criteria, UAT, SQL analytics, dashboards, and governance.

## **2. Business Problem**

The existing digital healthcare platform relies on multiple manual and fragmented processes across provider onboarding, verification, appointment management, consultation, payment, notifications, and operational support. Provider credential verification, document validation, compliance processing, and approval workflows require multiple handoffs, while downstream healthcare processes depend on successful provider activation. This can result in operational bottlenecks, delayed provider activation, inconsistent service levels, fragmented workflows, and limited visibility across the healthcare lifecycle.

### **Key challenges include:**

- Manual verification of provider credentials.
- Long onboarding turnaround times.
- Limited visibility into pending verification cases.
- High operational effort for compliance teams.
- Inconsistent SLA adherence.
- Delayed provider activation impacts business growth.
- Lack of centralized reporting for operational performance.
- Fragmented appointment and consultation workflows.
- Payment and notification dependencies across the healthcare journey.

- Limited end-to-end operational visibility.

These inefficiencies reduce operational productivity, affect provider and patient experience, and limit the platform's ability to scale its digital healthcare operations.

### **3. Business Objectives**

The project aims to modernize the provider onboarding journey through workflow automation and standardized operational processes.

#### **Primary Objectives**

- Digitize and standardize provider registration and onboarding workflows.
- Automate credential and regulatory verification wherever applicable.
- Improve provider approval, compliance processing and activation turnaround time.
- Enable standardized appointment and consultation workflows after provider activation.
- Support integrated payment and notification workflows across the healthcare journey.
- Provide real-time operational visibility through dashboards, SLA monitoring and KPI reporting.
- Improve auditability, access control and compliance tracking across critical workflows.
- Reduce manual operational effort and improve the overall provider and patient experience.

### **4. Project Scope**

#### **In Scope**

##### **Provider Lifecycle**

- Provider registration and profile creation
- Credential and document submission/validation
- Regulatory verification
- Automated approval/rejection and compliance exception handling
- Provider activation

##### **Patient & Healthcare Operation**

- Patient management
- Appointment management
- Consultation management
- Prescription/consultation completion workflow

- Payment processing and refund workflow
- Notifications
- Support ticket handling

## Operations & Governance

- SLA monitoring
- Operational dashboards and KPI reporting
- Audit logs
- Role-based access
- Compliance tracking

## Out of Scope

- Electronic Health Records (EHR)
- Pharmacy management
- Laboratory management
- Hospital administration
- Claims processing
- Payroll management
- Insurance reimbursement workflows

# Phase 2 – Business Analysis & Process Discovery

## 5. Stakeholder Analysis

### Stakeholder Matrix

| Stakeholder     | Role              | Key Responsibilities                                          | Primary Expectations                                              |
|-----------------|-------------------|---------------------------------------------------------------|-------------------------------------------------------------------|
| Patients        | End Users         | Book appointments, consult providers, make payments           | Fast access to verified providers and seamless booking experience |
| Providers       | Service Providers | Register, complete onboarding, deliver consultations          | Faster onboarding and timely activation                           |
| Operations Team | Operations        | Verify documents, manage onboarding queue, resolve exceptions | Reduced manual effort and improved SLA compliance                 |

|                  |                      |                                                                 |                                                            |
|------------------|----------------------|-----------------------------------------------------------------|------------------------------------------------------------|
| Compliance Team  | Regulatory Oversight | Validate licenses, ensure regulatory compliance, conduct audits | Accurate verification and complete audit trail             |
| Support Team     | Customer Support     | Resolve provider and patient issues, manage escalations         | Faster issue resolution and improved customer satisfaction |
| Product Team     | Product Management   | Define business requirements, prioritize enhancements           | Improved platform adoption and business outcomes           |
| Engineering Team | Development          | Build integrations, workflows and platform features             | Clear functional requirements and stable integrations      |

## Stakeholder RACI Matrix

| Activity              | Product | Operations | Compliance | Engineering | Support |
|-----------------------|---------|------------|------------|-------------|---------|
| Requirement Gathering | A       | C          | C          | C           | I       |
| Provider Verification | I       | R          | A          | I           | I       |
| API Integration       | C       | I          | C          | R/A         | I       |
| Dashboard Development | A       | C          | I          | R           | I       |
| UAT                   | A       | R          | C          | C           | R       |
| Production Deployment | C       | I          | I          | A/R         | C       |

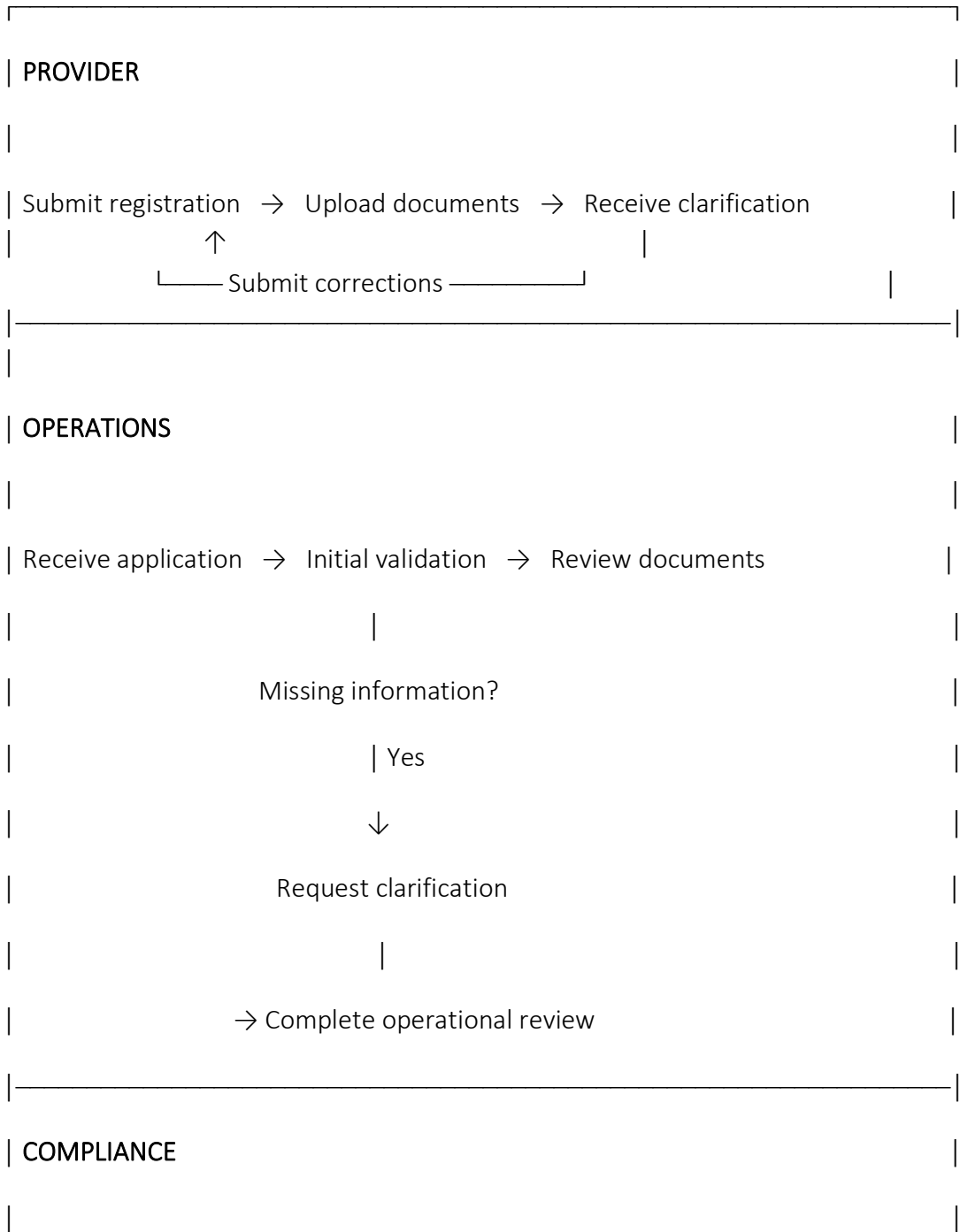
## Legend:

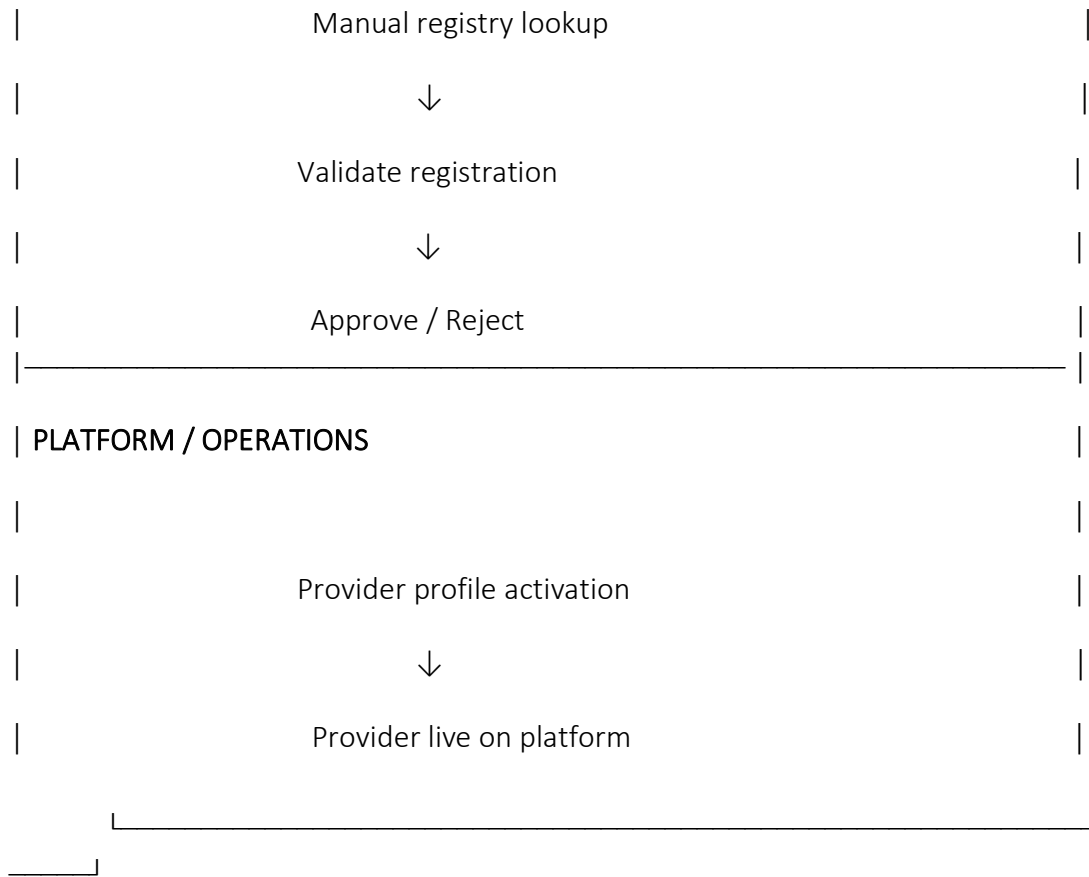
- R – Responsible
- A – Accountable
- C – Consulted
- I – Informed

## 6. Current Process (AS-IS)

### Current Workflow

The existing provider onboarding process relies heavily on manual activities across Operations and Compliance teams. Each onboarding request passes through multiple validation stages before a provider becomes active on the platform.





## Current Process

- The provider submits registration details.
- Supporting documents are uploaded.
- Operations perform initial validation.
- Compliance manually verifies medical registration.
- Missing information is communicated manually.
- The provider submits corrections.
- Operations complete review.
- Provider profile is activated.

Current Challenges

| Challenge                  | Business Impact                  |
|----------------------------|----------------------------------|
| Manual verification        | Longer onboarding time           |
| Multiple handoffs          | Operational delays               |
| No automated validation    | Higher error rate                |
| Lack of real-time tracking | Poor visibility                  |
| Manual communication       | Increased operational effort     |
| Limited dashboards         | Difficult performance monitoring |
| SLA breaches               | Reduced provider satisfaction    |

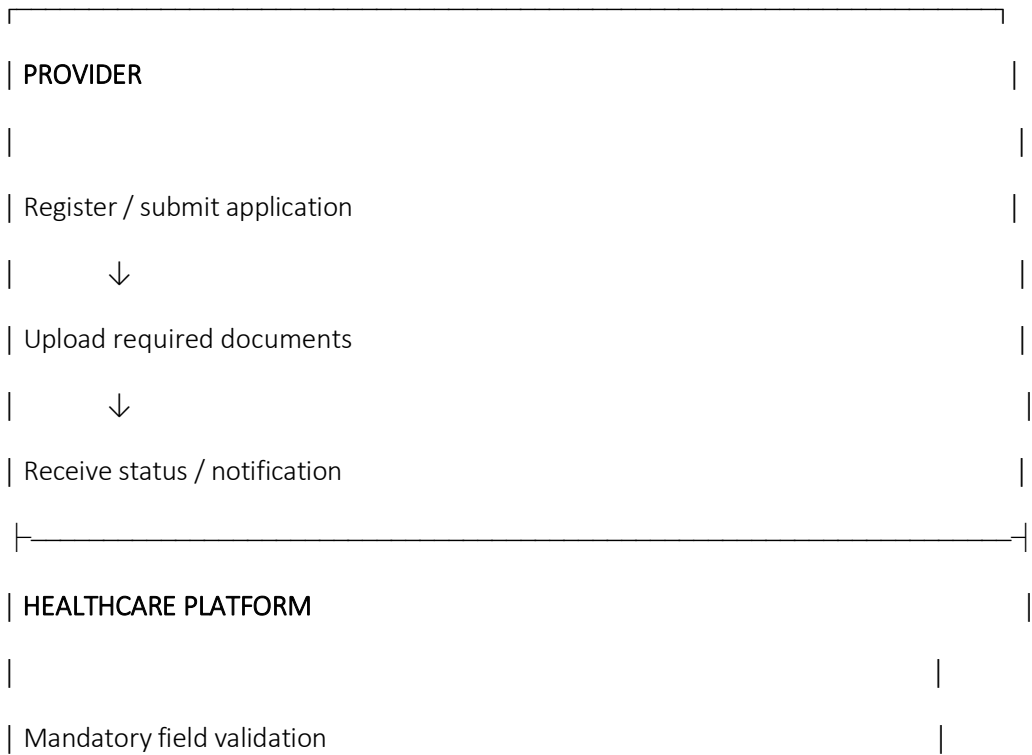
Reference Diagram

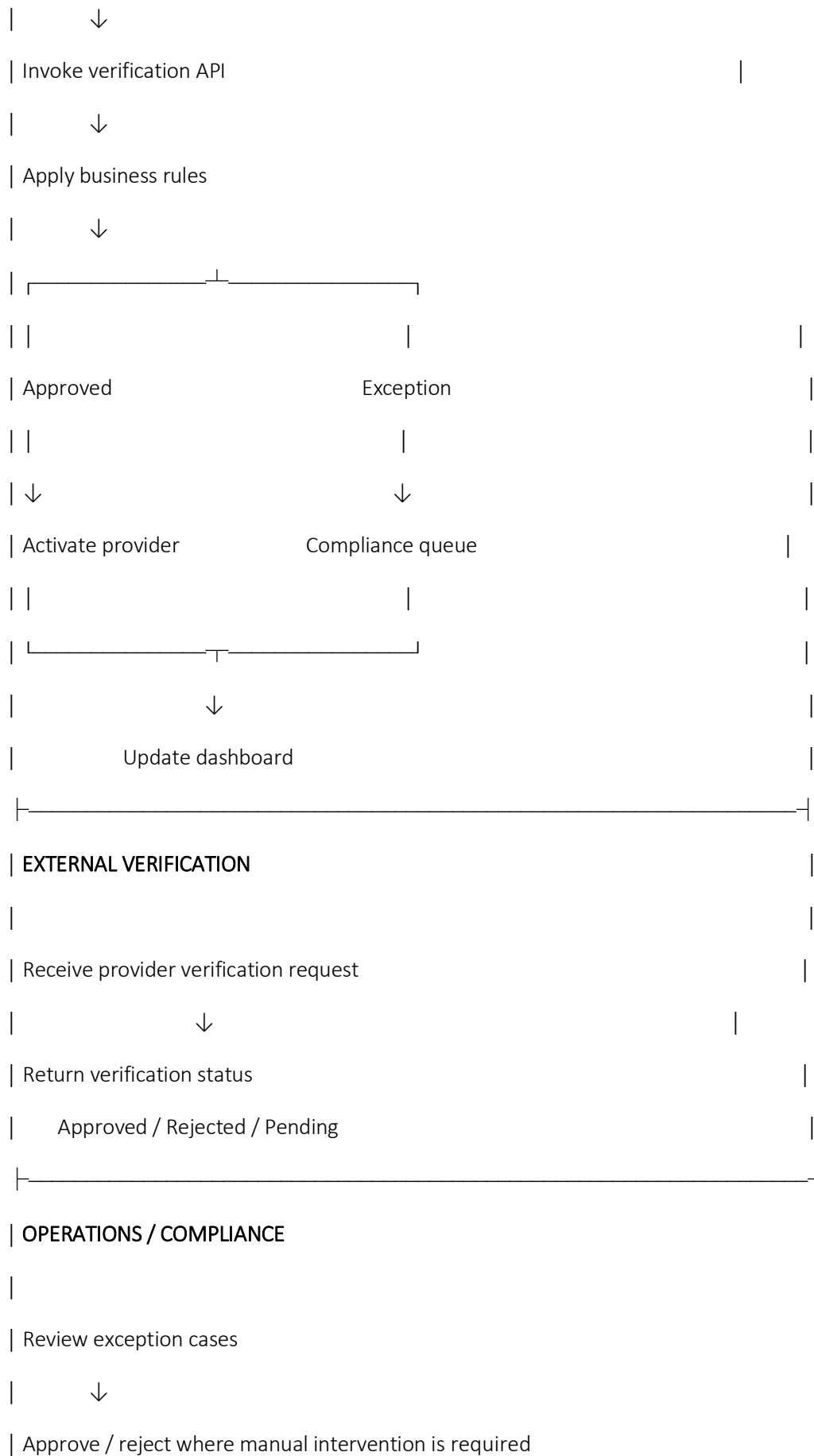
Refer to Asset 1 – Slide 16 (AS-IS Swimlane Process Diagram).

7. Future Process (TO-BE)

Proposed Workflow

The proposed onboarding solution introduces workflow automation, API-driven verification, business rules, and operational dashboards to reduce manual intervention while improving compliance.





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## Proposed Process

1. Provider registers.
2. Documents are uploaded.
3. System validates mandatory fields.
4. NMC Verification API triggered.
5. Automatic verification response received.
6. Business Rules Engine evaluates outcomes.
7. Low-risk providers auto-approved.
8. Exceptions routed to Compliance.
9. Notifications are sent automatically.
10. Provider profile is activated.
11. Dashboard updated in real time.

## Expected Improvements

| Current State        | Future State            |
|----------------------|-------------------------|
| Manual verification  | Automated verification  |
| Manual routing       | Rule-based routing      |
| Manual communication | Automated notifications |
| Static reporting     | Live dashboards         |
| Longer turnaround    | Reduced onboarding time |
| Limited visibility   | Real-time monitoring    |

## Business Benefits

- Reduced onboarding turnaround time.
- Improved SLA compliance.
- Faster provider activation.
- Better regulatory compliance.
- Reduced operational workload.
- Improved provider experience.
- Enhanced reporting and decision-making.

## Reference Diagram

*Refer to Asset 1 – Slide 17 (TO-BE Swimlane Process Diagram).*

## Phase 3 – Requirements Engineering

### 8. Business Requirements

#### Business Requirement Summary

| BR ID | Business Requirement                                                                | Business Objective                     |
|-------|-------------------------------------------------------------------------------------|----------------------------------------|
| BR-01 | Enable digital provider registration through a centralized onboarding portal.       | Improve provider onboarding experience |
| BR-02 | Automate provider for credential verification using external regulatory registries. | Reduce manual verification effort      |
| BR-03 | Standardize onboarding workflows with configurable business rules.                  | Improve process consistency            |
| BR-04 | Route verification exceptions to Compliance for manual review.                      | Ensure regulatory compliance           |
| BR-05 | Provide onboarding status updates to providers through automated notifications.     | Enhance user experience                |
| BR-06 | Enable automated notifications via Email/SMS/WhatsApp.                              | Improve communication efficiency       |
| BR-07 | Capture operational metrics for SLA and KPI reporting.                              | Support business decision-making       |
| BR-08 | Maintain complete audit logs for all onboarding activities.                         | Ensure compliance and traceability     |

### 9. Functional Requirements

#### Module 1 – Provider Registration

| FR ID | Functional Requirement |
|-------|------------------------|
|-------|------------------------|

|       |                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------|
| FR-01 | The system shall allow providers to register using personal, professional, and contact information. |
| FR-02 | The system shall validate all mandatory fields before submission.                                   |
| FR-03 | The system shall generate a unique Provider ID upon successful registration.                        |

## Module 2 – Provider Verification

| FR ID | Functional Requirement                                                                            |
|-------|---------------------------------------------------------------------------------------------------|
| FR-04 | The system shall automatically invoke the Provider Verification API upon profile submission.      |
| FR-05 | The system shall validate registration details against the National Medical Council for registry. |
| FR-06 | The system shall automatically approve providers with an Active license status.                   |
| FR-07 | The system shall route Suspended or Expired licenses to the Compliance queue for manual review.   |

## Module 3 – Appointment Activation

| FR ID | Functional Requirement                                                                          |
|-------|-------------------------------------------------------------------------------------------------|
| FR-08 | The system shall activate provider appointment scheduling after successful onboarding approval. |
| FR-09 | The system shall prevent appointment booking for inactive providers.                            |

## Module 4 – Notifications

| FR ID | Functional Requirement                                                                 |
|-------|----------------------------------------------------------------------------------------|
| FR-10 | The system shall send onboarding status notifications through Email, SMS, or WhatsApp. |
| FR-11 | The system shall notify Operations for pending verification beyond SLA thresholds.     |

## Module 5 – Dashboard & Reporting

| FR ID | Functional Requirement                                                                                    |
|-------|-----------------------------------------------------------------------------------------------------------|
| FR-12 | The system shall display onboarding KPIs in an executive dashboard.                                       |
| FR-13 | The system shall track Approval Rate, Rejection Rate, Pending Queue, Activation Rate, and SLA Compliance. |

## Module 6 – Payment & Consultation

| FR ID | Functional Requirement                                                                |
|-------|---------------------------------------------------------------------------------------|
| FR-14 | The system shall initiate payment processing for booked consultations.                |
| FR-15 | The system shall update payment status based on gateway response.                     |
| FR-16 | The system shall create consultation records after successful appointment completion. |

## Module 7 – Audit Trail

|       |                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------|
| FR-17 | The system shall record onboarding status changes and relevant user actions in an audit trail for authorized review. |
|-------|----------------------------------------------------------------------------------------------------------------------|

## 10. Non-Functional Requirements

| Category        | Requirement                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| Performance     | Verification responses should be processed within defined SLA limits where external services are available.     |
| Availability    | The platform should be available 24×7, excluding scheduled maintenance windows.                                 |
| Scalability     | The solution should support increasing provider onboarding volumes without significant performance degradation. |
| Reliability     | Failed API requests should be retried and logged for operational review.                                        |
| Security        | Sensitive provider and patient data should be encrypted in transit and at rest.                                 |
| Auditability    | All onboarding lifecycle events should be recorded for audit purposes.                                          |
| Usability       | The onboarding workflow should be intuitive and minimize manual effort.                                         |
| Maintainability | Business rules should be configurable with minimal development effort.                                          |

## 11. Business Rules

| Rule ID    | Business Rule                                                              |
|------------|----------------------------------------------------------------------------|
| BR-Rule-01 | A provider cannot be activated without successful regulatory verification. |

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| BR-Rule-02 | Providers with Suspended or Expired licenses must be routed to the Compliance queue.  |
| BR-Rule-03 | Appointment booking is permitted only by Active providers.                            |
| BR-Rule-04 | Payment must be successfully completed before appointment confirmation.               |
| BR-Rule-05 | Pending verification cases exceeding the defined SLA must be escalated to Operations. |
| BR-Rule-06 | Every onboarding status change must generate an audit log entry.                      |
| BR-Rule-07 | Only authorized roles can approve or reject provider applications.                    |
| BR-Rule-08 | Dashboard metrics must refresh based on the latest onboarding transaction data.       |

## Phase 4 – Functional Design

### 12. Functional Architecture

#### System Overview

The Healthcare Digital Platform is designed using a layered architecture to separate user interactions, business processes, integrations, and data management. This approach improves maintainability, scalability, and integration with external healthcare services.

#### High-Level Functional Architecture

| Layer              | Components                                                                                | Purpose                                  |
|--------------------|-------------------------------------------------------------------------------------------|------------------------------------------|
| Presentation Layer | Provider Portal, Patient Portal, Operations Dashboard, Compliance Dashboard, Admin Portal | User interaction and workflow management |

|                   |                                                                                                                                     |                                                |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Application Layer | Provider Onboarding Module, Appointment Management Module, Consultation Module, Payment Module, Notification Module, Support Module | Executes business processes and workflow logic |
| Integration Layer | Provider Verification API (NMC), Payment Gateway API, Notification API (Email/SMS/WhatsApp)                                         | Enables communication with external systems    |
| Data Layer        | Providers, Registries Master Patients Master, Appointments, Consultations, Payments, Support Tickets, Onboarding Transactions       | Stores operational and transactional data      |

## Functional Flow

1. Provider registers through the onboarding portal.
2. System validates mandatory information.
3. Provider Verification API validates registration details.
4. Business Rules Engine evaluates verification responses.
5. Provider is Approved, Rejected, or routed for Manual Review.
6. Notification service informs stakeholders.
7. Approved providers become available for appointments.
8. Payment and consultation workflows continue.
9. Dashboard updates operational KPIs.

## 13. Entity Relationship Model (ERD)

### Database Overview

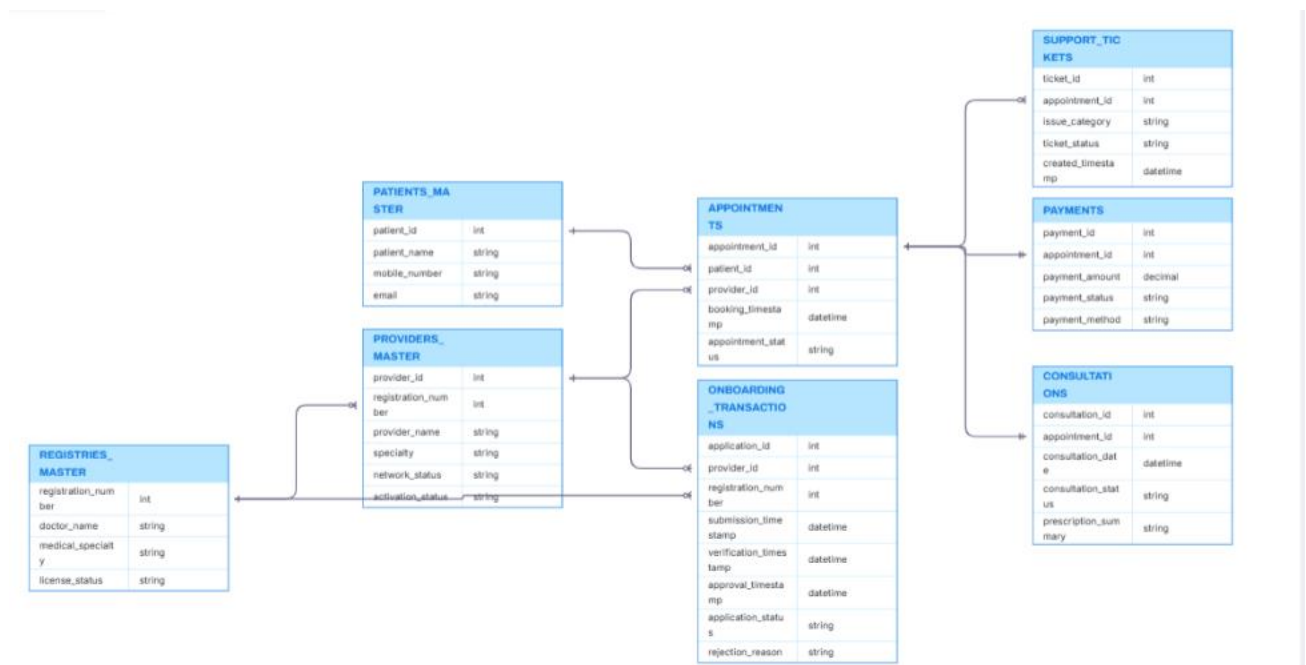
The platform uses eight primary entities to support onboarding, appointment management, consultation tracking, payments, customer support, and operational reporting.

Data Model Scope: The solution architecture defines an 8-entity conceptual data model covering the broader healthcare platform. The executable PostgreSQL schema focuses on the three core entities required for the provider onboarding and analytics use case: Registries Master, Providers Master, and Onboarding Transactions.

### Relationship Summary

| Parent Entity     | Relationship | Child Entity            |
|-------------------|--------------|-------------------------|
| Registries Master | 1 : Many     | Providers Master        |
| Providers Master  | 1 : Many     | Onboarding Transactions |
| Patients Master   | 1 : Many     | Appointments            |
| Providers Master  | 1 : Many     | Appointments            |
| Appointments      | 1 : 1        | Consultations           |
| Appointments      | 1 : 1        | Payments                |
| Appointments      | 1 : Many     | Support Tickets         |

## ERD Reference



## 14. Data Dictionary

Table 1 – Registries Master

| Field               | Data Type    | Description                        |
|---------------------|--------------|------------------------------------|
| registration_number | VARCHAR(30)  | Unique medical registration number |
| doctor_name         | VARCHAR(100) | Registered provider name           |
| medical_specialty   | VARCHAR(50)  | Clinical specialization            |

|                |             |                            |
|----------------|-------------|----------------------------|
| license_status | VARCHAR(20) | Active, Suspended, Expired |
|----------------|-------------|----------------------------|

**Table 2 – Providers Master**

| Field               | Data Type    | Description                 |
|---------------------|--------------|-----------------------------|
| provider_id         | INT          | Primary Key                 |
| registration_number | VARCHAR(30)  | Foreign Key                 |
| provider_name       | VARCHAR(100) | Provider name               |
| specialty           | VARCHAR(50)  | Medical specialty           |
| network_status      | VARCHAR(20)  | In-Network / Out-of-Network |
| activation_status   | VARCHAR(20)  | Active / Inactive           |

**Table 3 – Patients Master**

| Field         | Data Type    | Description    |
|---------------|--------------|----------------|
| patient_id    | INT          | Primary Key    |
| patient_name  | VARCHAR(100) | Patient name   |
| mobile_number | VARCHAR(15)  | Contact number |
| email         | VARCHAR(100) | Email address  |

**Table 4 – Onboarding Transactions**

| Field                  | Data Type   | Description            |
|------------------------|-------------|------------------------|
| application_id         | INT         | Primary Key            |
| provider_id            | INT         | Foreign Key            |
| registration_number    | VARCHAR(30) | Foreign Key            |
| submission_timestamp   | TIMESTAMP   | Registration submitted |
| verification_timestamp | TIMESTAMP   | Verification completed |
| approval_timestamp     | TIMESTAMP   | Approval completed     |

|                    |              |                               |
|--------------------|--------------|-------------------------------|
| application_status | VARCHAR(20)  | Approved / Rejected / Pending |
| rejection_reason   | VARCHAR(100) | Reason for rejection          |

**Table 5 – Appointments**

| Field              | Data Type   | Description                       |
|--------------------|-------------|-----------------------------------|
| appointment_id     | INT         | Primary Key                       |
| patient_id         | INT         | Foreign Key                       |
| provider_id        | INT         | Foreign Key                       |
| booking_timestamp  | TIMESTAMP   | Appointment booking time          |
| appointment_status | VARCHAR(20) | Scheduled / Completed / Cancelled |

**Table 6 – Consultations**

| Field                | Data Type   | Description           |
|----------------------|-------------|-----------------------|
| consultation_id      | INT         | Primary Key           |
| appointment_id       | INT         | Foreign Key           |
| consultation_date    | TIMESTAMP   | Consultation date     |
| consultation_status  | VARCHAR(20) | Completed / Cancelled |
| prescription_summary | TEXT        | Consultation summary  |

**Table 7 – Payments**

| Field          | Data Type     | Description                 |
|----------------|---------------|-----------------------------|
| payment_id     | INT           | Primary Key                 |
| appointment_id | INT           | Foreign Key                 |
| payment_amount | DECIMAL(10,2) | Consultation amount         |
| payment_status | VARCHAR(20)   | Success / Failed / Refunded |
| payment_method | VARCHAR(30)   | UPI / Card / Net Banking    |

Table 8 – Support Tickets

| Field             | Data Type   | Description                                  |
|-------------------|-------------|----------------------------------------------|
| ticket_id         | INT         | Primary Key                                  |
| appointment_id    | INT         | Foreign Key                                  |
| issue_category    | VARCHAR(50) | Payment Failure, SLA Breach, Technical Issue |
| ticket_status     | VARCHAR(20) | Open / In Progress / Closed                  |
| created_timestamp | TIMESTAMP   | Ticket creation time                         |

### Data Flow Summary

| Source                    | Destination             | Purpose               |
|---------------------------|-------------------------|-----------------------|
| Provider Portal           | Providers Master        | Provider registration |
| Provider Verification API | Registries Master       | License verification  |
| Onboarding Module         | Onboarding Transactions | Workflow tracking     |
| Appointment Module        | Appointments            | Booking management    |
| Consultation Module       | Consultations           | Consultation records  |
| Payment Gateway           | Payments                | Payment tracking      |
| Support Module            | Support Tickets         | Issue management      |
| Analytics Engine          | Dashboard               | KPI reporting         |

## Phase 5 – Agile Backlog & User Stories

### 15. Agile Product Backlog

#### Epic 1 – Provider Registration & Onboarding

Enable healthcare providers to complete a secure, digital onboarding process with minimal manual intervention.

#### US-HEALTH-01: Provider Registration

As a healthcare provider

I want to register using my professional and personal information

So that I can initiate the onboarding process.

##### Acceptance Criteria (Gherkin)

Given the provider is on the registration page

When all mandatory details are entered

Then the system shall create a provider profile

And generate a unique Provider ID.

#### US-HEALTH-02: Document Upload

As a healthcare provider

I want to upload mandatory verification documents

So that my profile can be verified.

##### Acceptance Criteria

Given the provider has completed registration

When required documents are uploaded

Then the application shall allow profile

submission. US-HEALTH-03: Profile Validation

As an Operations Executive

I want mandatory field validation before submission

So that incomplete applications do not enter the verification queue.

##### Acceptance Criteria

Given mandatory information is missing

When the provider clicks Submit

Then the application shall display validation errors

And prevent submission.

## Epic 2 – Provider Verification & Compliance

Automate provider verification while ensuring regulatory compliance.

#### US-HEALTH-04: Provider Verification API

As an Operations Executive

I want provider credentials verified automatically

So that manual verification effort is reduced.

##### Acceptance Criteria

Given onboarding has been submitted

When the system invokes the NMC Verification API

Then the provider status shall be updated based on the API

response. US-HEALTH-05: Auto Approval

As a Compliance Manager

I want providers with valid licenses to be automatically approved

So that onboarding time is reduced.

##### Acceptance Criteria

Given the API returns an Active license

When verification completes

Then the application shall set the provider status to Approved

And activate the provider profile.

#### US-HEALTH-06: Compliance Queue

As a Compliance Executive

I want rejected or suspicious applications routed for manual review

So that regulatory risks are minimized.

##### Acceptance Criteria

Given verification fails

When the provider has an Expired or Suspended license

Then the application shall move the request to the Compliance Queue.

### **Epic 3 – Appointment & Consultation Management**

Enable verified providers to manage appointments and complete consultations through a

structured workflow.

#### US-HEALTH-07: Appointment Activation

As a verified healthcare provider

I want my appointment calendar to become available immediately after successful onboarding  
So that patients can start booking consultations.

##### Acceptance Criteria

Given the provider's onboarding status is Approved

When the activation process completes

Then the provider profile shall be marked as Active

And appointment slots shall become available for booking.

#### US-HEALTH-08: Consultation Completion

As a healthcare provider

I want to mark consultations as completed

So that consultation records are maintained accurately.

##### Acceptance Criteria

Given an appointment is in Scheduled status

When the provider clicks Complete Consultation

Then the appointment status shall change to Completed

And a consultation record shall be created.

#### US-HEALTH-09: Prescription Upload

As a healthcare provider

I want to upload consultation notes and prescriptions

So that patients can securely access their treatment information.

##### Acceptance Criteria

Given a consultation is completed

When the provider uploads prescription details

Then the prescription shall be linked to the consultation record

And be accessible to the patient.

## Epic 4 – Payments & Notifications

Enable secure payment processing and automated communication throughout the provider and patient journey.

### US-HEALTH-10: Payment Processing

As a patient

I want to complete payment securely before my consultation

So that my appointment is confirmed.

#### Acceptance Criteria

Given a patient proceeds to checkout

When payment is successfully authorized

Then the payment status shall become Success

And the appointment status shall change to

Scheduled. US-HEALTH-11: Automated Notifications

As a provider and patient

I want to receive automated onboarding and appointment notifications

So that I remain informed about important status updates.

#### Acceptance Criteria

Given an onboarding or appointment status changes

When the event is processed

Then the system shall send Email, SMS, or WhatsApp notifications

And record the notification status.

### US-HEALTH-12: Refund Processing

As a patient

I want to receive an automatic refund if my appointment is cancelled

So that I do not need to manually request reimbursement.

#### Acceptance Criteria

Given an appointment is cancelled

When payment has already been completed

Then the system shall initiate the refund process  
And update the payment status to Refunded.

## **Epic 5 – Operations & Reporting**

Provide operational visibility, SLA monitoring, and proactive issue management through dashboards and support workflows.

### US-HEALTH-13: SLA Monitoring

As an Operations Manager  
I want to monitor onboarding requests approaching SLA deadlines  
So that delays can be addressed proactively.

#### Acceptance Criteria

Given an onboarding request exceeds the defined SLA threshold  
When the dashboard refreshes  
Then the application shall highlight the request  
And notify the Operations team.

### US-HEALTH-14: Executive Dashboard

As a Business Manager  
I want to view operational KPIs through an executive dashboard  
So that I can monitor business performance and make informed decisions.

#### Acceptance Criteria

Given dashboard data is refreshed  
When the manager accesses the dashboard  
Then KPIs including Approval Rate, Rejection Rate, Pending Queue, Activation Rate, and SLA Compliance shall be displayed.

### US-HEALTH-15: Support Ticket Escalation

As a Support Executive  
I want failed onboarding, payment, or system issues to automatically generate support tickets So that customer issues are resolved efficiently.

### Acceptance Criteria

Given a payment failure, API failure, or onboarding exception occurs  
When the system detects the event  
Then a support ticket shall be created automatically  
And assigned to the appropriate support queue.

### US-HEALTH-16: Audit Trail

As a Compliance Manager,  
I want all onboarding status changes and actions to be recorded in an audit trail,  
so that onboarding activity can be traced for compliance and review

### Acceptance Criteria

Given an onboarding application exists  
when the onboarding status changes  
Then the system shall create an audit log entry for the status change  
And the audit trail shall be available for authorized review

## 16. Agile Release Planning

| Sprint   | Features                              |
|----------|---------------------------------------|
| Sprint 1 | Registration, Login, Document Upload  |
| Sprint 2 | Verification API, Compliance Workflow |
| Sprint 3 | Appointment & Consultation            |
| Sprint 4 | Payments & Notifications              |
| Sprint 5 | Dashboard, Reporting & UAT            |

### Definition of Ready (DoR)

A user story is considered ready when:

- Business objective is defined.
- Acceptance criteria are documented.
- Dependencies are identified.

- UI/Process flow is available (if applicable).
- Stakeholder approval is obtained.

## Definition of Done (DoD)

A user story is considered complete when:

- Functional requirements are implemented.
- Acceptance criteria are satisfied.
- UAT is successfully completed.
- No critical defects remain.
- Business stakeholder approves the feature.

## Phase 6 – Integration Specifications

### 17. Integration Overview

#### Purpose

The Healthcare Digital Platform integrates with external services to automate provider verification, payment processing, and stakeholder notifications. These integrations reduce manual effort, improve processing speed, and ensure seamless communication between systems.

| Integration               | Purpose                                                           | Trigger                                 | Response                                            |
|---------------------------|-------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------|
| Provider Verification API | Validate provider credentials with National Medical Council (NMC) | Provider submits onboarding application | Verification status (Approved / Rejected / Pending) |
| Payment Gateway API       | Process consultation payments                                     | Patient confirms appointment            | Payment Success / Failure / Refunded                |

|                  |                                       |                               |                       |
|------------------|---------------------------------------|-------------------------------|-----------------------|
| Notification API | Send Email/SMS/WhatsApp notifications | Status change or system event | Delivery confirmation |
|------------------|---------------------------------------|-------------------------------|-----------------------|

## 18. API Contract 1 – Provider Verification API

### Business Purpose

Automatically validate provider registration details against the National Medical Council (NMC) registry before onboarding approval.

### API Details

| Attribute      | Value                         |
|----------------|-------------------------------|
| Method         | POST                          |
| Endpoint       | <code>/provider/verify</code> |
| Authentication | OAuth 2.0 Bearer Token        |
| Content Type   | application/json              |

### Sample Request

```
{  
  "registration_number": "MH12345", "provider_name": "Dr. Priya Sharma",  
  "state_council": "Maharashtra Medical Council"  
}
```

### Sample Success Response

```
{  
  "status": "SUCCESS", "license_status": "Active",  
  "verification_result": "Verified", "action": "AUTO_APPROVE"  
}
```

### Sample Failure Response

```
{  
  
"status": "FAILED",  
"license_status": "Suspended",  
"verification_result": "Not Verified",  
"action": "MANUAL_REVIEW",  
"reason": "License Suspended"  
  
}
```

## Business Rules

- Only Active licenses qualify for automatic approval.
- Suspended or Expired licenses are routed to the Compliance queue.
- All verification attempts are logged for audit purposes

## 19. API Contract 2 – Payment Gateway API

### Business Purpose

Enable secure payment processing for patient consultations.

### API Details

| Attribute      | Value             |
|----------------|-------------------|
| Method         | POST              |
| Endpoint       | /payments/process |
| Authentication | API Key           |
| Content Type   | application/json  |

### Sample Request

```
{  
  
"appointment_id": 5001,  
"patient_id": 301,  
"provider_id": 101,
```

```
"amount": 1500, "currency": "INR"
```

```
}
```

Sample Success Response

```
{
```

```
"status": "SUCCESS",  
"payment_reference": "PAY123456",  
"payment_status": "Completed"
```

```
}
```

Sample Failure Response

```
{
```

```
"status": "FAILED",  
"payment_status": "Declined",  
"reason": "Insufficient Balance"
```

```
}
```

## Business Rules

- Appointment confirmation occurs only after successful payment.
- Failed payments allow retry without duplicate booking creation.
- Refunds are initiated automatically for cancelled appointments (where applicable).

## 20. API Contract 3 – Notification API

### Business Purpose

Notify providers and patients about onboarding, appointments, payments, and operational updates.

| Attribute | Value               |
|-----------|---------------------|
| Method    | POST                |
| Endpoint  | /notifications/send |

|                |                      |
|----------------|----------------------|
| Authentication | API Key              |
| Channels       | Email, SMS, WhatsApp |

#### Sample Request

```
{
  "recipient": "+919999999999",
  "channel": "WhatsApp",
  "template": "Provider_Approved",
  "provider_name": "Dr. Priya Sharma"
}
```

#### Sample Success Response

```
{
  "status": "DELIVERED",
  "message_id": "MSG458923"
}
```

#### Sample Failure Response

```
{
  "status": "FAILED",
  "reason": "Invalid Mobile Number"
}
```

### Business Rules

- Notifications are triggered after onboarding, payment, and appointment events.
- Delivery failures are logged for operational review.
- Multiple channels (Email, SMS, WhatsApp) can be configured based on user preference.

### Integration Sequence

#### Provider

|



Provider Portal

|



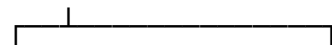
Verification API

|



Business Rules Engine

|



|

Approved      Manual Review

|

|



Notification      Compliance Queue

|



Provider Activated

## Integration Error Handling

| Scenario                 | System Action                                                                     |
|--------------------------|-----------------------------------------------------------------------------------|
| Verification API timeout | Retry request; if unsuccessful, move application to Pending and notify Operations |

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Payment failure      | Keep appointment unconfirmed and allow payment retry              |
| Notification failure | Log the failure and retry through an alternate configured channel |

## Phase 7 – Testing & Validation

### 21. User Acceptance Testing (UAT)

#### UAT Objective

The objective of User Acceptance Testing (UAT) is to validate that the Healthcare Digital Platform meets business requirements and is ready for production deployment. UAT is performed by business users to ensure the solution supports operational workflows, regulatory requirements, and user expectations.

#### UAT Entry Criteria

- Functional development completed.
- System Integration Testing (SIT) completed successfully.
- Test environment available.
- Test data prepared.
- Business requirements approved.
- Test cases reviewed and signed off.

#### UAT Test Cases

| UAT ID | Feature               | Test Scenario                              | Expected Result                       | Status  |
|--------|-----------------------|--------------------------------------------|---------------------------------------|---------|
| UAT-01 | Provider Registration | Register a new provider with valid details | Provider profile created successfully | Planned |
| UAT-02 | Document Upload       | Upload mandatory verification documents    | Documents uploaded successfully       | Planned |

|        |                        |                                                                     |                                                                                                                 |         |
|--------|------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------|
| UAT-03 | Provider Verification  | Verify an Active provider via NMC API                               | Provider automatically approved                                                                                 | Planned |
| UAT-04 | Compliance Queue       | Verify a Suspended provider                                         | Application routed to Compliance                                                                                | Planned |
| UAT-05 | Appointment Activation | Activate appointment after approval                                 | Provider becomes available for booking                                                                          | Planned |
| UAT-06 | Payment Processing     | Complete payment successfully                                       | Payment status updated to Success                                                                               | Planned |
| UAT-07 | Notification Service   | Send onboarding approval notification                               | WhatsApp/Email delivered                                                                                        | Planned |
| UAT-08 | Dashboard              | Verify KPI dashboard updates                                        | Latest onboarding metrics displayed                                                                             | Planned |
| UAT-09 | SLA Monitoring         | Delay onboarding beyond SLA                                         | Escalation alert generated                                                                                      | Planned |
| UAT-10 | Support Ticket         | Trigger payment failure                                             | Support ticket created automatically                                                                            | Planned |
| UAT-11 | Audit Trail            | Verify that onboarding status changes and user actions are recorded | All relevant onboarding actions are captured in the audit trail with user, action, timestamp and status details | Planned |

## UAT Exit Criteria

- All critical business scenarios executed.
- No Critical or High severity defects remain open.
- Business stakeholders approve the solution.
- Functional requirements validated.
- UAT Sign-off pending.

## 22. Requirements Traceability Matrix

| Business Requirement | Functional Requirement | User Story   | UAT    |
|----------------------|------------------------|--------------|--------|
| BR-01                | FR-01, FR-02           | US-01        | UAT-01 |
| BR-02                | FR-04, FR-05           | US-04        | UAT-03 |
| BR-03                | FR-06, FR-07           | US-05, US-06 | UAT-04 |
| BR-04                | FR-07                  | US-06        | UAT-04 |
| BR-05                | FR-10                  | US-11        | UAT-07 |
| BR-06                | FR-10, FR-11           | US-11        | UAT-07 |
| BR-07                | FR-12, FR-13           | US-14        | UAT-08 |
| BR-08                | FR-17                  | US-16        | UAT-11 |

## UAT Stakeholders

| Stakeholder      | Responsibility                                               |
|------------------|--------------------------------------------------------------|
| Product Manager  | Validate business objectives                                 |
| Business Analyst | Prepare test cases, coordinate UAT, support issue resolution |
| Operations Team  | Validate onboarding workflow                                 |
| Compliance Team  | Verify regulatory scenarios                                  |
| QA Team          | Support defect validation                                    |
| Engineering Team | Fix identified defects                                       |
| Business Sponsor | Final business sign-off                                      |

Defect Severity Matrix

| Severity | Description                        | Expected Resolution |
|----------|------------------------------------|---------------------|
| Critical | Core business function unavailable | Immediate           |
| High     | Major functionality impacted       | Before release      |
| Medium   | Partial functionality affected     | Planned fix         |
| Low      | Cosmetic or minor usability issue  | Future release      |

UAT Sign-off

| Role               | Sign-off Status |
|--------------------|-----------------|
| Product Manager    | Not Executed    |
| Business Analyst   | Not Executed    |
| Operations Manager | Not Executed    |
| Compliance Manager | Not Executed    |

Sections in Phase 8

23. Data Analytics Overview

The Healthcare Digital Platform captures provider onboarding lifecycle data to monitor operational workload, processing performance, SLA compliance, and exception patterns. The illustrative analytics dataset contains 20 onboarding transactions across provider and registry master data. PostgreSQL SQL queries transform the transaction data into operational KPIs and analytical outputs, which are visualized through a Tableau Operations Dashboard.

24. SQL Analytics

| SQL Analysis                        | Business Purpose                                                  |
|-------------------------------------|-------------------------------------------------------------------|
| SLA Compliance & Average Turnaround | Measure onboarding processing performance against the 24-hour SLA |

|                                     |                                                                   |
|-------------------------------------|-------------------------------------------------------------------|
| Rejection Reason Ranking            | Identify the most frequent onboarding rejection bottlenecks       |
| Specialty-level Onboarding Activity | Analyze sequential onboarding activity across medical specialties |

**SQL techniques demonstrated:** CTEs, conditional aggregation, `NULLIF`, `DENSE_RANK()`, `LAG()`, joins, grouping, and timestamp-based calculations.

## 25. KPI Framework

| KPI                   | Formula / Calculation                                                 | Current Illustrative Result |
|-----------------------|-----------------------------------------------------------------------|-----------------------------|
| Total Applications    | Count of onboarding transactions                                      | 20                          |
| Approved Applications | Count where status = Approved                                         | 12                          |
| Rejected Applications | Count where status = Rejected                                         | 4                           |
| Pending Applications  | Count where status = Pending                                          | 4                           |
| Approval Rate         | Approved / Total Applications                                         | 60%                         |
| SLA Compliance        | Approved applications completed $\leq 24$ hrs / Approved applications | 58.33%                      |
| Average Turnaround    | Average of Approval Timestamp – Submission Timestamp                  | 28.79 hrs                   |
| Top Rejection Reason  | Highest rejection count                                               | Document Mismatch — 2 cases |

## 26. Dashboard Overview

- Total Applications
- Approved Applications
- Rejected Applications
- Pending Applications

- SLA Compliance
- Average Turnaround
- Application Status Distribution
- Rejection Reason Analysis

## 27. Business Insights

### Insight 1 — SLA Performance

Average onboarding turnaround is 28.79 hours, while 58.33% of approved applications meet the 24-hour SLA threshold.

Recommendation:

Prioritize SLA-breaching cases through automated alerts and escalation workflows.

### Insight 2 — Rejection Bottleneck

Document Mismatch is the leading rejection reason, accounting for 2 of 4 rejected applications.

Recommendation:

Strengthen document validation and completeness checks during registration.

### Insight 3 — Current Application Pipeline

The illustrative dataset contains 4 pending applications alongside 12 approved and 4 rejected applications.

Recommendation:

Use queue monitoring and SLA-based escalation to manage pending cases.

### Insight 4 — Specialty-level Analysis

The SQL analysis tracks sequential onboarding activity within each medical specialty using `LAG()`, enabling analysis of submission patterns across specialties.

Recommendation:

Use specialty-level operational analysis to identify workload concentration and support resource planning.

**Decision Matrix.**

| Observation                | Recommendation       | Expected Business Impact     |
|----------------------------|----------------------|------------------------------|
| High Pending Queue         | Auto-escalation      | Faster approvals             |
| Frequent Document Mismatch | Front-end validation | Lower rejection rate         |
| SLA Breaches               | Priority routing     | Better compliance            |
| Payment Failures           | Retry workflow       | Improved customer experience |

## Phase 9 – Governance, Compliance & Risk Management

### 28. Governance & Compliance

#### Objective

Ensure the provider onboarding process complies with regulatory requirements, protects sensitive data, and maintains complete auditability throughout the onboarding lifecycle.

#### Regulatory Compliance

| Regulation                  | Application in Project                                |
|-----------------------------|-------------------------------------------------------|
| DPDP Act 2023               | Protect provider and patient personal information     |
| Healthcare Data Privacy     | Restrict unauthorized access to sensitive information |
| Internal Audit Policy       | Maintain complete onboarding audit logs               |
| Information Security Policy | Enforce secure handling of provider credentials       |

#### Role-Based Access Control

| Role     | Access Level                     |
|----------|----------------------------------|
| Provider | Manage own profile and documents |

|                      |                                                  |
|----------------------|--------------------------------------------------|
| Operations Team      | Review onboarding applications                   |
| Compliance Team      | Verify credentials and approve/reject exceptions |
| Support Team         | Manage customer issues and escalations           |
| Product Team         | View dashboards and reports                      |
| System Administrator | Full platform access                             |

## Audit & Logging

The system maintains a complete audit trail for every onboarding transaction, including:

- Provider registration
- Document submission
- Verification status changes
- Approval/Rejection actions
- Payment status updates
- Notification history
- User activity logs

## 29. Assumptions, Dependencies & Constraints

### Assumptions

- Providers submit accurate registration details.
- External verification services are available.
- Required documents are uploaded during registration.
- Users have internet connectivity.

### Dependencies

- National Medical Council Verification API
- Payment Gateway
- Notification Service
- Internal Operations Team
- Compliance Team

### Constraints

- Third-party API availability
- Regulatory compliance requirements

- Network latency
- Data privacy policies
- SLA commitments

### 30. Risk Register

| Risk                           | Probability | Impact | Mitigation                                 |
|--------------------------------|-------------|--------|--------------------------------------------|
| NMC API downtime               | Medium      | High   | Retry mechanism and manual verification    |
| Incorrect provider information | Medium      | High   | Mandatory validation and compliance review |
| Payment failures               | Medium      | Medium | Retry workflow and support escalation      |
| SLA breaches                   | Medium      | High   | Automated alerts and priority queues       |
| Unauthorized data access       | Low         | High   | RBAC and encryption                        |
| Notification delivery failure  | Low         | Medium | Retry through alternate channels           |

### 31. Assumptions & Decision Log

| Decision                          | Reason                                    |
|-----------------------------------|-------------------------------------------|
| Automated verification introduced | Reduce manual effort                      |
| Dashboard implemented             | Improve operational visibility            |
| Rule-based approval               | Standardize decision making               |
| Payment module included           | Support end-to-end consultation lifecycle |

|                         |                                   |
|-------------------------|-----------------------------------|
| Notification automation | Improve stakeholder communication |
|-------------------------|-----------------------------------|

### 32. Security Considerations

| Area              | Control                                       |
|-------------------|-----------------------------------------------|
| Authentication    | Secure user login                             |
| Authorization     | Role-Based Access Control (RBAC)              |
| Data Encryption   | Encrypt sensitive data in transit and at rest |
| Audit Logging     | Record all business-critical actions          |
| API Security      | Token-based authentication                    |
| Password Security | Encrypted password storage                    |

#### Governance Checklist

| Item                  | Status    |
|-----------------------|-----------|
| Data Privacy          | Addressed |
| Audit Logging         | Addressed |
| Access Control        | Addressed |
| Regulatory Compliance | Addressed |
| Risk Management       | Addressed |
| Security Controls     | Addressed |

## Phase 10 – Project Closure & Business Outcomes

### 33. Business Impact Assessment

#### Expected Business Benefits

|               |                  |
|---------------|------------------|
| Business Area | Expected Outcome |
|---------------|------------------|

|                        |                                                                           |
|------------------------|---------------------------------------------------------------------------|
| Provider Onboarding    | Reduced onboarding turnaround time through workflow automation            |
| Operational Efficiency | Reduced manual verification effort and standardized processes             |
| Regulatory Compliance  | Improved compliance through automated verification and audit trails       |
| Provider Experience    | Faster onboarding and real-time application status updates                |
| Customer Support       | Reduced support effort through automated notifications and issue tracking |
| Executive Reporting    | Improved operational visibility through dashboards and KPIs               |

### Key Performance Indicators (KPIs)

| KPI                        | Baseline                       | Target                     |
|----------------------------|--------------------------------|----------------------------|
| Average Onboarding Time    | 5 Business Days                | < 24 Hours                 |
| Provider Approval Rate     | 60% (illustrative dataset)     | > 80%                      |
| Manual Verification Effort | High                           | Reduced through automation |
| SLA Compliance             | 70%(Assumed Business Baseline) | > 95%                      |
| Dashboard Availability     | Not Available                  | Real-Time Visibility       |

**Note:** The 70% SLA compliance figure represents the assumed current-state business baseline used for this portfolio case study. SQL and dashboard metrics are calculated independently from the illustrative sample dataset and are not production measurements.

## 34. Project Deliverables

| Deliverable                          | Status    |
|--------------------------------------|-----------|
| Business Requirements Document (BRD) | Completed |
| Functional Requirements              | Completed |

|                                 |           |
|---------------------------------|-----------|
| Non-Functional Requirements     | Completed |
| Business Rules                  | Completed |
| Functional Architecture         | Completed |
| ERD & Data Dictionary           | Completed |
| API Contracts                   | Completed |
| Agile Backlog (16 User Stories) | Completed |
| Gherkin Acceptance Criteria     | Completed |
| UAT Test Pack                   | Prepared  |
| SQL Analytics                   | Completed |
| Tableau Dashboard               | Completed |
| Governance & Compliance         | Completed |

## 35. Lessons Learned

### Key Learnings

- Early stakeholder alignment reduces requirement changes.
- Automating verification improves operational efficiency.
- Standardized business rules improve process consistency.
- API integrations reduce manual dependency and turnaround time.
- Executive dashboards enable faster, data-driven decision-making.
- Well-defined UAT scenarios improve release quality and business confidence.

## 36. Future Enhancements

| Enhancement                      | Business Value                   |
|----------------------------------|----------------------------------|
| AI-assisted document validation  | Reduce manual document review    |
| OCR-based certificate extraction | Faster provider onboarding       |
| Multi-registry verification      | Broader regulatory coverage      |
| Mobile provider onboarding       | Improved accessibility           |
| Predictive SLA risk alerts       | Proactive operational management |

|                                 |                          |
|---------------------------------|--------------------------|
| Advanced operational dashboards | Deeper business insights |
|---------------------------------|--------------------------|

## 37. Appendix

### Supporting Assets

- Healthcare Digital Platform PPT (Asset 1)
- Business Analysis Documentation (Asset 2)
- SQL Database Schema (schema.sql)
- SQL Analytics Queries (analytics\_queries.sql)
- Sample Dataset
- Tableau Dashboard
- ERD Diagram
- AS-IS Swimlane Diagram
- TO-BE Swimlane Diagram
- API Specifications

### Tools Used

| Category      | Tool                 |
|---------------|----------------------|
| Documentation | Microsoft Word       |
| Presentation  | Microsoft PowerPoint |
| Database      | PostgreSQL           |
| SQL           | PostgreSQL SQL       |
| Dashboard     | Tableau Public       |

### Document Summary

**Project Name:** Healthcare Digital Platform

**Domain:** HealthTech | Provider Lifecycle | Patient Care | Digital Healthcare Operations

**Role Demonstrated:** Business Analyst

**Methodology:** Agile

## Key Capabilities Demonstrated:

- Business Analysis
- Requirements Engineering
- Process Mapping
- Functional Design
- API Analysis
- Data Modeling
- SQL Analytics
- Dashboard Design
- User Story Writing
- UAT Planning
- Governance & Compliance
- Stakeholder Management











































